

ASSIGNMENT REPORT-5

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For The Subject

Pattern Recognition and Machine Learning

(26AI5707)



DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AND DATA SCIENCE

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INTRODUCTION

This assignment studies Maximum Likelihood Estimation (MLE) for Gaussian data. In Question 1, three two-dimensional classes are generated with different covariance structures. The data is split into training and test sets, class means and covariance matrices are estimated from training data, and the test samples are classified. Decision-region plots, accuracy and confusion matrices are used to evaluate the classifier. In Question 2, the same Gaussian classification approach is applied to the supplied real dataset. The objectives for this experiment as follow

- Estimate the Gaussian class means and covariance matrices with MLE.
- Compare the results with isotropic, diagonal and full covariance matrices.
- Study shared covariance and class specific covariance models.
- Classify unseen data and obtain the accuracy and the confusion matrix.
- Plot the decision regions and the ROC curves.

EXPERIMENTAL SETUP

Item	Description
Synthetic classes	3 two-dimensional Gaussian classes
Samples generated	10,000 per class; 30,000 total
Synthetic split	80% training (24,000), 20% testing (6,000)
Class means	(0, 0), (5, 5), (0, 7)
Covariance types	Isotropic, diagonal and full
Covariance assumptions	Shared across classes or different per class
Real training file	trian(1).txt: 1,050 samples; 350 per class
Real development file	dev(1).txt: 300 samples; 100 per class
Libraries	NumPy and Matplotlib

THEORY AND FORMULAS

For class k with N_k training samples $\mathbf{x}_i$, the MLE mean is:

$$\hat{\boldsymbol{\mu}}_k = (1/N_k) \sum \mathbf{x}_i$$

and the MLE estimate of the covariance matrix is:

$$\hat{\boldsymbol{\Sigma}}_k = (1/N_k) \sum (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^T$$

In general, for each class a Gaussian score is calculated, and the sample is allocated to the most likely class based on the maximum score. The score is implemented using the log Gaussian density to avoid numerical underflow due to such huge exponents.

If the covariance matrix is shared then quadratic terms cancel when the class scores are compared and the decision boundary between any two classes is linear. On the other hand, if the classes have independent covariance matrices then quadratic terms do not cancel and the decision boundary is not linear.

EXPERIMENT 1 - SYNTHETIC GAUSSIAN DATA

For each experiment, 10,000 points are generated for each of the three classes. An 80:20 split is made separately within each class. The means and covariance matrices are estimated from the training samples, and the test samples are classified using Gaussian scores.

Experiment	Test accuracy
Isotropic – shared covariance	99.75%
Diagonal – shared covariance	93.53%
Full – shared covariance	97.30%
Isotropic – different covariance	96.93%
Diagonal – different covariance	95.27%
Full – different covariance	95.22%

DECISION – BOUNDARY PLOTS

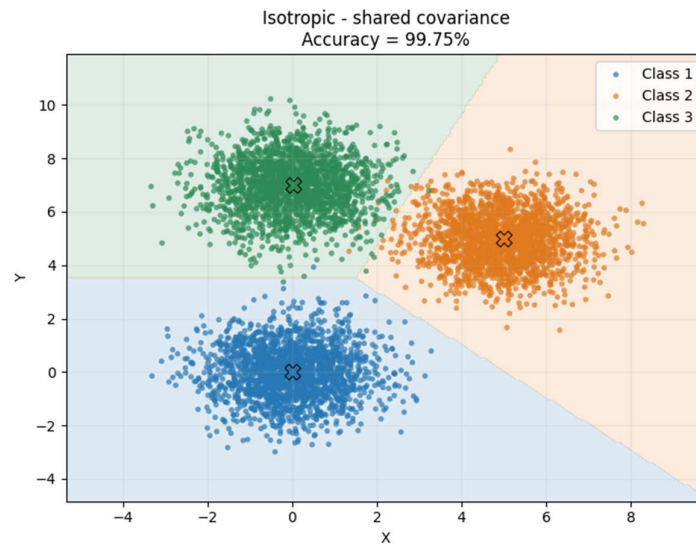


Figure 1. Isotropic covariance shared across classes.

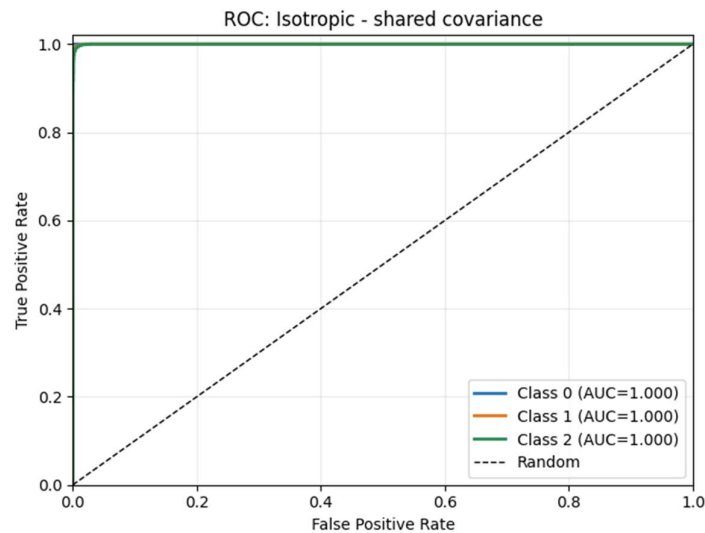


Figure 2: ROC Plot with Shared Isotropic Covariance

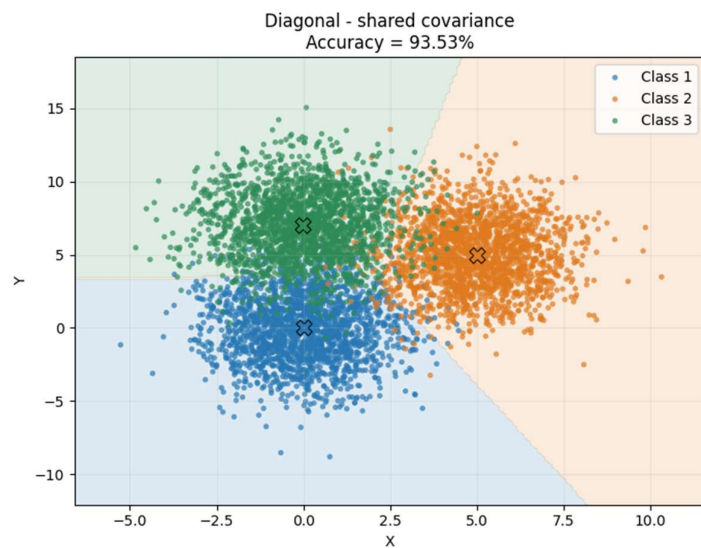


Figure 3. Diagonal covariance shared across classes

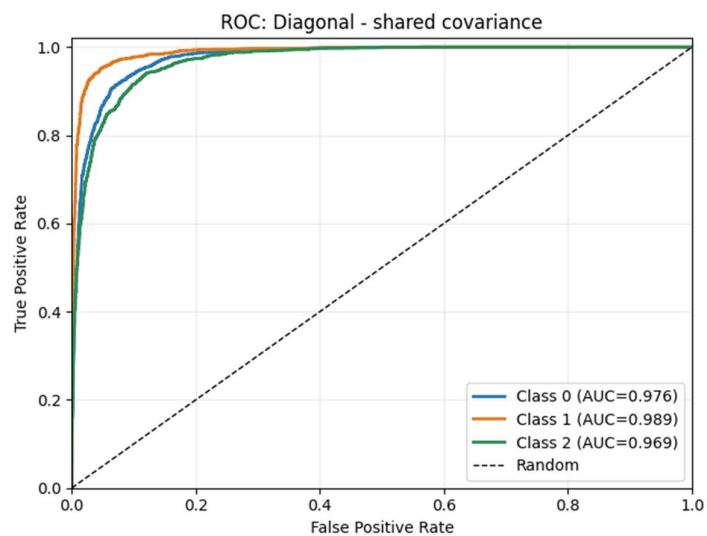


Figure 4: ROC Curve – Diagonal Shared Isotropic Covariances

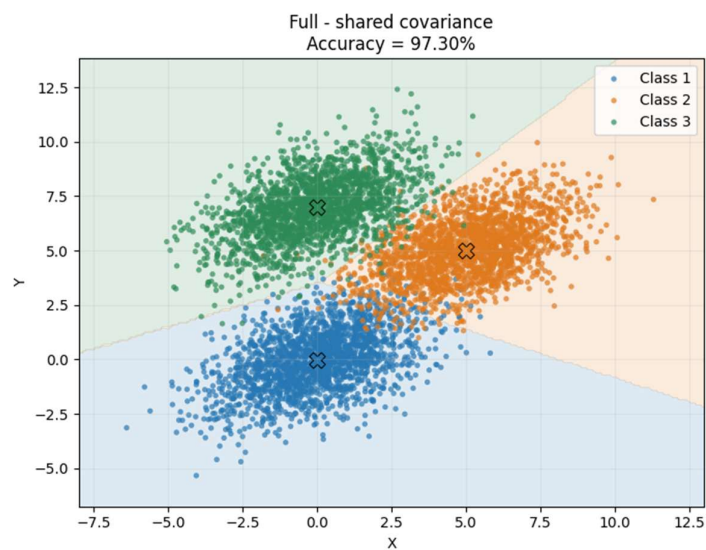


Figure 5. Full covariance shared across classes.

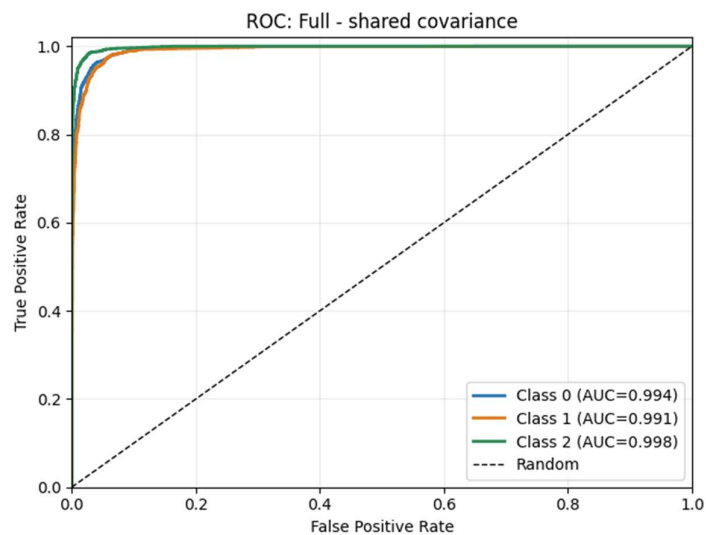


Figure 6. ROC Curve – Full Shared Isotropic Covariance

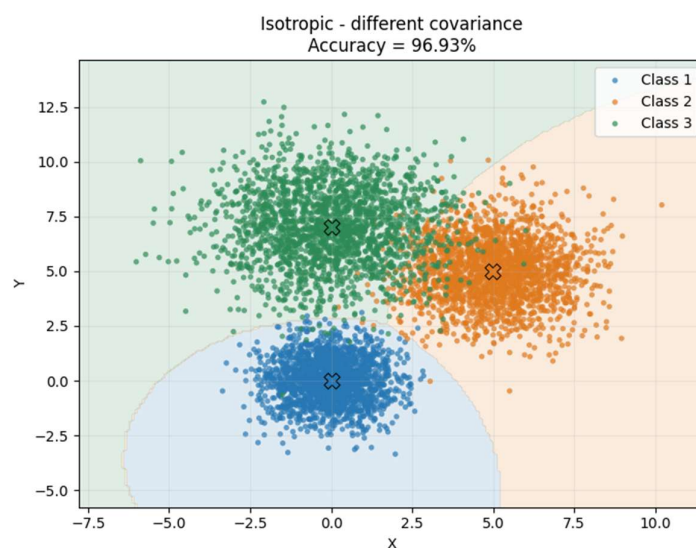


Figure 7. Isotropic covariance differs by class

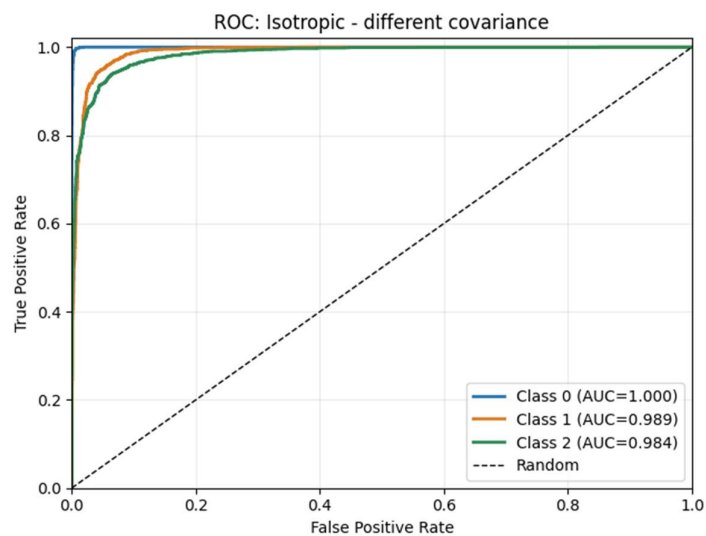


Figure 8. ROC Curve – Isotropic with different covariance

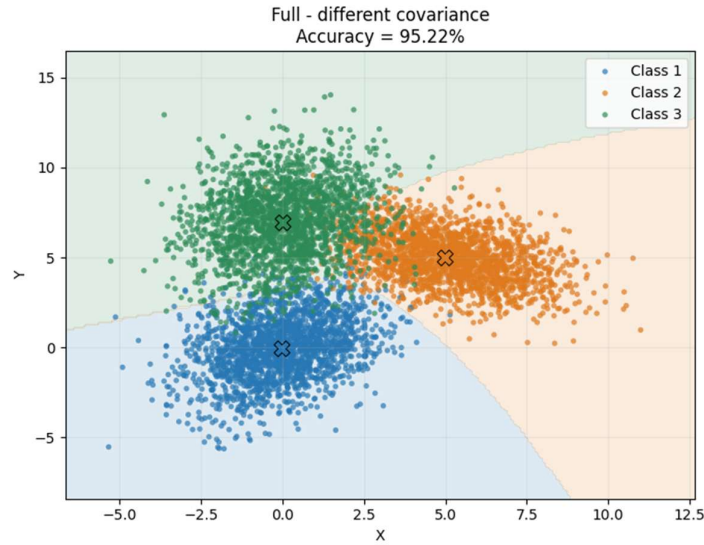


Figure 9. Full covariance differs by class

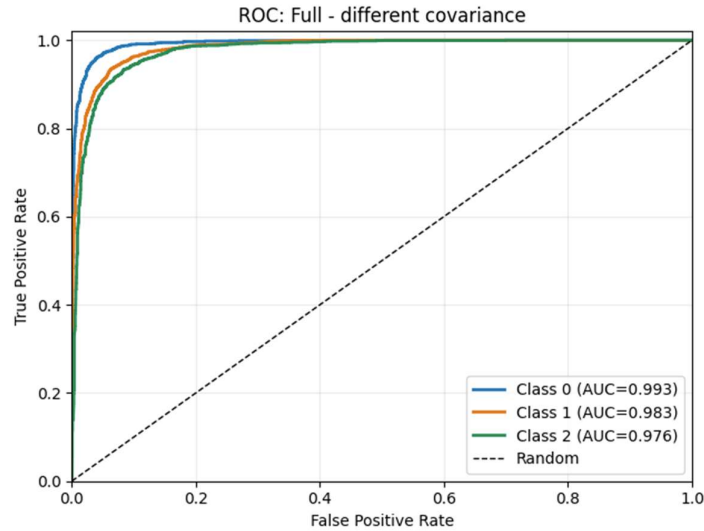


Figure 10. ROC Curve- Full different Isotropic Covariance

OBSERVATION

- Isotropic shared-covariance experiment gave the highest accuracy (99.75%) for this generated data set.
- Diagonal shared-covariance experiment had the lowest accuracy (93.53%) suggesting that there is more overlap between classes for this type of covariance.
- Different covariance matrices allow for different spreads and orientations for each class, leading to non-linear decision boundaries.
- Misclassifications tend to occur where classes overlap.

EXPERIMENT 2 - SUPPLIED REAL DATASET

The given data file includes three attributes; namely, the x coordinate, y coordinate, and the class attribute. Training data include 1,050 samples where each class is represented by 350 samples while development data include 300 samples where each class is represented by 100 samples. Parameters including the means and class-specific covariances are calculated based on the given training data.

ESTIMATED MEANS

Class	Mean X	Mean Y
1	583.756	1440.500
2	409.563	1899.483
3	409.069	1205.462

CLASSIFICATION RESULT

- Accuracy on the 300 development samples: 81.00%.
- Confusion matrix (rows = actual class; columns = predicted class)

Actual \ Predicted	Class 1	Class 2	Class 3
Class 1	81	9	10
Class 2	10	89	1
Class 3	24	3	73

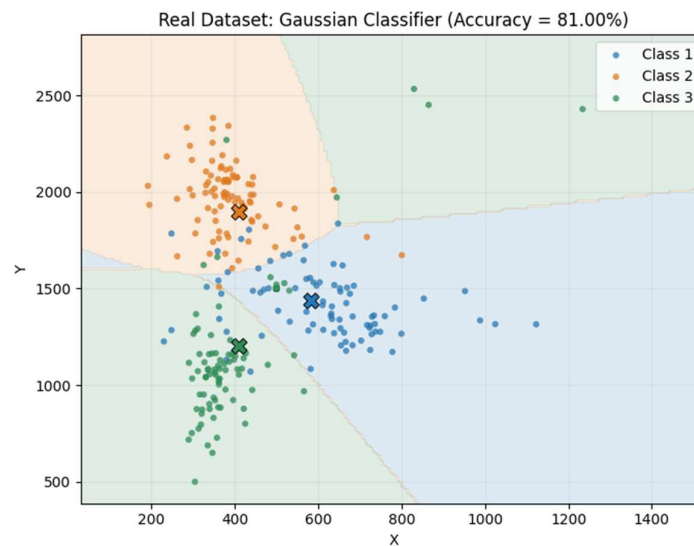


Figure 11. Decision regions and development samples for the real dataset.

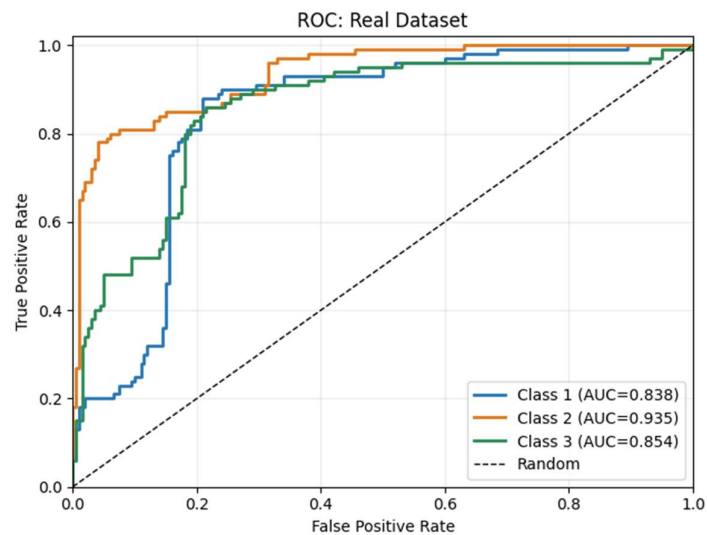


Figure 12. One-vs-rest ROC curves for the real dataset.

OBSERVATION

- The classifier correctly classified 243 of 300 development samples, giving 81.00% accuracy.
- Class 1 was correctly classified 81 times, Class 2 was correctly classified 89 times, and Class 3 was correctly classified 73 times.
- The main errors occur between Class 1 and Class 3, and between Class 2 and Class 1.
- The real data is less cleanly separated than the synthetic Gaussian data, so its accuracy is lower.

CONCLUSION

This project highlighted the capabilities of the maximum likelihood estimation method in estimating the mean and covariances of the Gaussian classes and their application in classifying new instances. It was revealed that the covariance structure influences the form of the decision boundaries and the precision of the classification process. The synthetic dataset was able to yield accuracies ranging between 93.53% and 99.75%, whereas the provided real dataset was able to achieve 81.00% accuracy.